SMART RESTAURANT MANAGEMENT SYSTEM

Core Modules

Functional modules, features and dependencies

DOCUMENT ID	SRMS-03
VERSION	1.0
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STATUS	Draft for review
PHASE	Phase 1: event-driven core | Phase 2: AI camera detection
PURPOSE	Breaks the system into modules with features, inputs, outputs and dependencies.

1. Module Map

The system is divided into functional modules. Phase 1 modules are the MVP; Module M14 is added in Phase 2.

ID	Module	Purpose	Phase
M1	QR and Table Session	Identify the table, open and manage the session	1
M2	Digital Menu	Present items, variants, modifiers, availability	1
M3	Cart and Ordering	Shared cart, place order, rounds, order history	1
M4	Kitchen Display (KDS)	Tickets, preparation status, timing	1
M5	Billing and Payments	Bill, tax, UPI/card/cash, refunds	1
M6	Table Management and Live Floor Map	Table states, floor map, manual override	1
M7	Alerts and Attention Engine	Raise, route and resolve table alerts	1
M8	Cleaning and Task Management	Cleaning queue, staff tasks	1
M9	Authentication and Staff	Users, roles, shifts, PIN login	1
M10	Admin and Configuration	Menu, tables, QR codes, taxes, thresholds	1
M11	Real-time Notifications	WebSocket events, push, sound	1
M12	Reports and Analytics	Sales, turnover, dwell, peak hours	1 basic, 3 advanced
M13	Feedback and Reservations/Waitlist	Ratings, comments, waiting list, bookings	1.5
M14	AI Camera Vision	Detect seating, clearing, queues and attention needs	2

M1. QR and Table Session

Entry point for every guest interaction.

- Each table has a unique QR encoding a random **table token** (not the table number alone, to prevent guessing).
- Scan opens the menu in the browser and creates or joins the active session for that table.
- If a session is active, new scanners join it (shared cart). If it has been closed, a new one starts.
- Optional guest count and name prompt.
- Session expiry on payment, idle timeout or manager closure.
- Admin can regenerate or print QR codes (PDF sheet with table labels).

Input: Table token **Output:** Session ID, table context

M2. Digital Menu

Everything the guest sees before ordering.

- Categories, items with photo, description, price, veg/non-veg marker, tags, prep time.
- Variants (half/full), modifier groups (extra cheese, spice level) with min/max rules.
- Search, filters, popular and recommended sections.
- Real-time availability: sold-out items are greyed out instantly.
- Multi-language support (English, Hindi, Gujarati) in a later release.

Input: Menu data **Output:** Menu JSON for UI

M3. Cart and Ordering

Turns browsing into a validated order.

- Shared cart per session with live sync between guests at the same table.
- Special instructions per item and per order.
- Price, tax and total are calculated on the server, never trusted from the client.

- Multiple order rounds in one session; each round is a separate order under one bill.
- Place order triggers kitchen ticket and waiter notification; optional waiter confirmation mode.
- Cancellation rule: allowed until kitchen accepts; later needs staff approval.

Input: Cart, session **Output:** Order, order items

M4. Kitchen Display (KDS)

Accurate tickets for the kitchen.

- Tickets appear in real time, oldest first, with table, items, modifiers and notes.
- Item status: Pending, Accepted, Preparing, Ready.
- Colour timers turn amber and red when an item exceeds its target prep time.
- Mark item out of stock; menu updates immediately.
- Optional station routing (grill, bar, dessert) in a later release.

Input: Orders **Output:** Item status events

M5. Billing and Payments

Fair, correct and fast payment.

- One running bill per session combining all orders.
- GST breakdown, service charge toggle, discounts (manager approved), tips.
- Online payment via UPI intent/QR, cards and wallets through a payment gateway (Razorpay or equivalent).
- Cash: waiter records receipt; bill closes.
- Idempotent payment handling and webhook verification to avoid double charge or missed payment.
- Partial payment and split bill (equal split first, item split later).
- Refunds with reason and approval; PDF/WhatsApp-ready receipt.

Input: Bill **Output:** Payment, receipt

M6. Table Management and Live Floor Map

The manager's main screen. Single source of truth for table status.

- Table states: **Available, Reserved, Occupied, Payment Pending, Needs Cleaning, Cleaning In Progress, Out of Service**.
- **Needs Attention** is an overlay flag from open alerts, not a separate state.
- Drag-and-drop floor layout editor in admin; multiple floors or zones.
- Each table tile shows label, state colour, guest count, session timer and alert icon.
- Side panel: orders, bill status, alerts and action buttons (seat, clean, override, close).
- Header KPIs: tables occupied, guests seated, capacity %, average dwell time, active alerts.
- Every state change is logged with actor and reason.

Input: Events from M1, M3, M5, M7, M8 **Output:** Table status, live map

M7. Alerts and Attention Engine

Makes sure no table is ignored.

- Alert types: Call Waiter, Request Bill, Order Delayed, Payment Failed, Long Wait (no order after seating), Unattended Table, Long Stay, Help.
- Rules and thresholds are configurable per restaurant (for example no order within 10 minutes of seating).
- Alert lifecycle: Open, Acknowledged (by whom), Resolved. Unacknowledged alerts escalate to the manager.
- Routing to the assigned waiter first, manager as fallback.
- Visual and sound cues on the floor map and staff phones.

Input: Events and timers **Output:** Alerts, escalations

M8. Cleaning and Task Management

Closes the loop from guest leaving to table ready.

- When payment completes (or the manager closes the table), a cleaning task is created and the table becomes Needs Cleaning.
- Cleaners see a prioritised queue (oldest first, reserved tables boosted).
- Start and Done actions with timestamps; average clean time tracked.
- Optional checklist (wipe, reset cutlery, sanitise).
- Manager can assign a task to a specific cleaner or let it be first-come.

Input: Table state **Output:** Cleaning tasks, table released

M9. Authentication and Staff

Secure identity for staff and sessions.

- Staff CRUD, role assignment, active/inactive.
- PIN login for floor staff; email/password plus optional 2FA for manager/owner.
- Shifts and table assignment per waiter.
- Audit trail of sensitive actions.

Input: Credentials **Output:** Tokens, permissions

M10. Admin and Configuration

Everything the owner configures once and tweaks later.

- Menu management with bulk import (CSV), images, availability schedule.
- Floors, tables, capacity, layout and QR generation.
- Tax rates, service charge, currency, receipt header.
- Alert thresholds and session timeouts.
- Payment gateway keys (owner only, stored encrypted).

Input: Admin input **Output:** Configuration

M11. Real-time Notifications

Live updates everywhere.

- WebSocket channels per restaurant (and per table for guests) for table, order, alert and payment events.
- Browser push and in-app sound for staff devices.
- Graceful fallback to polling if the socket drops; automatic resync on reconnect.

Input: Domain events **Output:** Live UI updates

M12. Reports and Analytics

Turns operations into decisions.

- Sales by day, hour, category, item; payment method split.
- Table turnover, average dwell time, average cleaning time.
- Alert response time, order preparation time.
- Peak-hour heat map by table and time.
- Export to CSV/PDF.

Input: All data **Output:** Dashboards, exports

M13. Feedback, Reservations and Waitlist

Experience and demand management.

- Post-payment rating and comment; low ratings alert the manager.
- Waitlist: add party, estimate wait, notify by SMS/WhatsApp when a table frees.

- Reservations: hold a table for a time window; the table shows Reserved on the map.

Input: Guest input **Output:** Feedback, queue

M14. AI Camera Vision (Phase 2)

Automatic table awareness from CCTV, as demonstrated in the video.

- Map each camera view to table zones; detect people, chairs and tables with an object detection model.
- Infer: guests seated, guests left, table cleared, staff near table, long unattended table, queue at entrance.
- Send events to the same state engine with a confidence score. High confidence updates automatically; low confidence becomes a suggestion for staff to confirm.
- Processing at the edge where possible; no face recognition or identity storage; store only counts, states and short event snapshots if enabled.
- Manual override always wins and feeds back to improve thresholds.

Input: Video streams **Output:** Vision events

2. Module Dependencies

Module	Depends on	Used by
M1 QR/Session	M10 (tables and QR)	M2, M3, M5, M6
M3 Ordering	M1, M2	M4, M5, M6
M5 Billing	M3	M6, M8, M12
M6 Floor map	M1, M3, M5, M7, M8	Manager UI, M12
M7 Alerts	M1, M3, M5, timers	M6, M11
M8 Cleaning	M5, M6	M6, M12
M14 AI vision	M6, M7, M10	M6, M7, M12